Capital One | Design Pro Bono

Nightingale Caring Solutions

Final readout

May 29, 2026



Confidential

Usability Study Synthesis

Categorizing feedback from users to clarify pain points.

Strength
themes

Managing the Staffing Roster

Adding Nurses
Both participants found the process of adding a new nurse to the team highly intuitive. Dorothy identified the "plus" sign immediately, and Heather quickly located it and successfully added her name to the roster.

Managing Shift Schedules
Populating and modifying shifts via drag-and-drop or direct selection was a breeze. Both participants successfully figured out how to drag a nurse's name directly into the current shift box without hesitation.

Removing Staff via Hover
Both users handled the task rapidly using the minus icon.

Understanding Core Conceptual Frameworks

Grabbing the Concept of "Factors"
The fundamental concept of "Factors" (patient attributes/precautions) was extremely well-received and understood right away by both participants. Dorothy easily listed what existing factors meant (such as fall risks or isolation needs) and noted they are very helpful for making fair assignments. Heather immediately recognized them as safety-critical patient attributes that play a major role in workload balance.

Creating Custom Factors
Adding a brand new item to the factor list was a seamless workflow. Dorothy successfully typed a factor into the box and clicked add, while Heather smoothly added "airborne precautions" to the environment.

Navigating Geographic Pods
Setting up physical clusters of rooms was highly intuitive. Both participants easily understood that "Pods" represented a set of geographically located rooms. Dorothy successfully used a "Add Pod" button and dragged a room down into it, while Heather quickly spun up "Pod E" and dragged a room in to test the geographic clustering.

Embracing Core Automation Features

The Value of Automated Shift "Assigning"
While the wayfinding and mathematical logic under the hood wasn't entirely clear to them, both participants successfully triggered the "Assign" functionality and immediately recognized its power. Dorothy called the auto-generation "easy" and "quick", while Heather remarked that auto-generating assignments with this information would be incredibly helpful because doing it manually in the real world is incredibly time-consuming.

Opportunity themes

Navigation and Discovery

Adding Rooms
Both participants struggled to find the specific "plus" signs or buttons required to add new rooms to the system.

Assignments Tab
Participants did not immediately notice the "Assignments" tab and instead defaulted to manual drag-and-drop or clicking into individual room numbers to assign staff.

Sidebar vs. Dashboard
The existence of both a sidebar and a bottom dashboard caused confusion regarding where specific actions should be performed.

Mental Model and Terminology Mismatches

Patient Centric View
Both users noted that they are used to seeing patient names rather than just room numbers; viewing only room numbers felt unintuitive and "dehumanizing".

Factor v. Workload score
While "Factors" (patient attributes) were well-understood, the "Workload Score" logic was confusing. Participants expected it to be an automatic calculation rather than something they had to manage manually.

Fast v. Analytical
Participants could not immediately distinguish between the "Fast" and "Analytical" assignment modes.

Technical and Visual UI Issues

Status indicators
The "C" for Charge Nurse was not easily seen or understood.

Feature clarity
The purpose of "View Pods" was not immediately clear to the participants.

Logic and Data Integration

Assignment math
The logic behind how the platform automatically distributed rooms and workload was not fully understood, especially when scores varied significantly between nurses.

Dynamic data
There was confusion about how factors would stay tied to patients if those patients were moved to different rooms, as the current system seemed to tie data to the room itself.

Full task breakdown

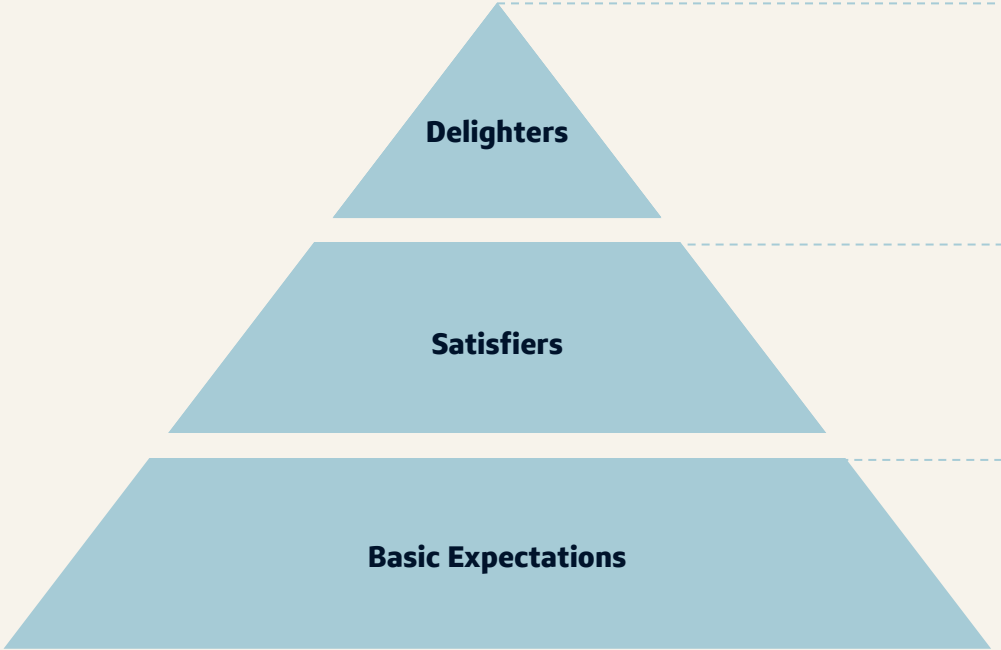
Successfully completed (independently) | Struggled | Could not complete without moderator help

	Add nurse to team	Add nurse to current shift	Remove nurse from current shift	Assign nurse to rooms	Remove room assignment from nurse	Add room	Remove room
Dorothy							
Heather							
	Set status of room to Discharged	**Clear all nurses from next shift**	**Clear all rooms from next shift**	**Add a factor**	**Add a workload score**	**Add a pod**	
Dorothy		N/A	N/A				
Heather							
	Assign a charge nurse	**Finds "Assignments" tab**	**Finds "Quick Actions"**	**Able to assign “1:1 Report”**	**Able to assign “Continuity”**	**Able to make an assignment via the "Assignments dashboard"**	
Dorothy							
Heather							

Prioritization of design improvements

An execution plan that plots design improvements into a clear, phased approach for Nightingale's development.

Framing product improvements to narrow focus and drive prioritization.



Example: Implement a shift-end feedback loop to help the algorithm learn what a "fair" assignment actually feels like. For example, a quick "rate this assignment" questionnaire for staff nurses.

Example: Implement a constraint-based patient reporting experience so there's a centralized place to log "radar patients" or specific concerns that the next shift charge nurse needs to see immediately

Example: Provide constraint-based matching by hard-coding safety rules (e.g., "pregnant nurse cannot take shingles patients") to guarantee the #1 goal of patient/nurse safety

Prioritize foundational stability within the basic expectations bucket to build immediate user trust by focusing on the high impact areas first.

High Impact

Foundations: Critical deal-breakers affecting visibility and user control.

- User control and freedom
- Flexibility and efficiency of use
- Consistency and standards
- Visibility of system status

Medium Impact

Workflows: Structural alignment with real-world clinical nursing patterns.

- Match between system and the real world
- Help users recognize, diagnose, and recover from errors

Low Impact

Details: Cumulative friction points that contribute to daily fatigue.

- Help and documentation
- Error prevention
- Consistency and standards
- Visibility of system status

Strategic Consult

Core areas of focus from the usability research and heuristic evaluation, accompanied with design recommendations.

Targeting key areas of user friction to deliver an immediate usability lift.

Object Relationships

Improving usability when adding an object (e.g. such as adding a room) and connection objects together in relation to the other (e.g. adding a nurse to a room)

Buttons

Improving the overall button system (e.g. primary, secondary, and tertiary) and as well as interactions (e.g. active, hover, disabled states) to improve findability throughout the app.

User Knowledge

Systematic improvements to help users understand the purpose of certain tools better (e.g. adding a workload score).

Hidden UI

Tackling the key accessibility and findability of items that are currently hidden in the UI and require hovering (e.g. assigning a charge nurse).

Standardize nurse and room creation patterns to reduce cognitive load and improve workflow speed.

ISSUE

Users are having difficulty adding a room because the experience is different and because the Rooms tab is hidden from the bottom panel on first view.

RECOMMENDATION

Add the add room button to the left room grid so they can add a room from there. Whether they are adding a room from the bottom panel or the left panel, the room should populate where they are currently at.

Change icons to labeled buttons when possible to provide a better understanding of what action is being done.

ISSUE

The plus and minus icons are used in different ways - the plus is to add rooms/nurses and the minus collapses the panel views and removes rooms/nurses.

RECOMMENDATION

Instead of having the plus icons for adding a room and nurse, make them buttons that say “Add room” or “Add room.” This increases the click target and also makes it more clear what the action is.

Change icons to labeled buttons and standardize nurse and room creation patterns



Provide guidance to what the Assignments tab and the different features can be used for.

ISSUE

The users have to make assumptions on what each feature in the Assignments does before clicking Assign.

RECOMMENDATION

Provide some instructions to what the Assignments tab is for and how they can use each feature underneath the tab to help them in making their assignments.

Make the Assignments tab and button clearer to see when first viewing the main page.

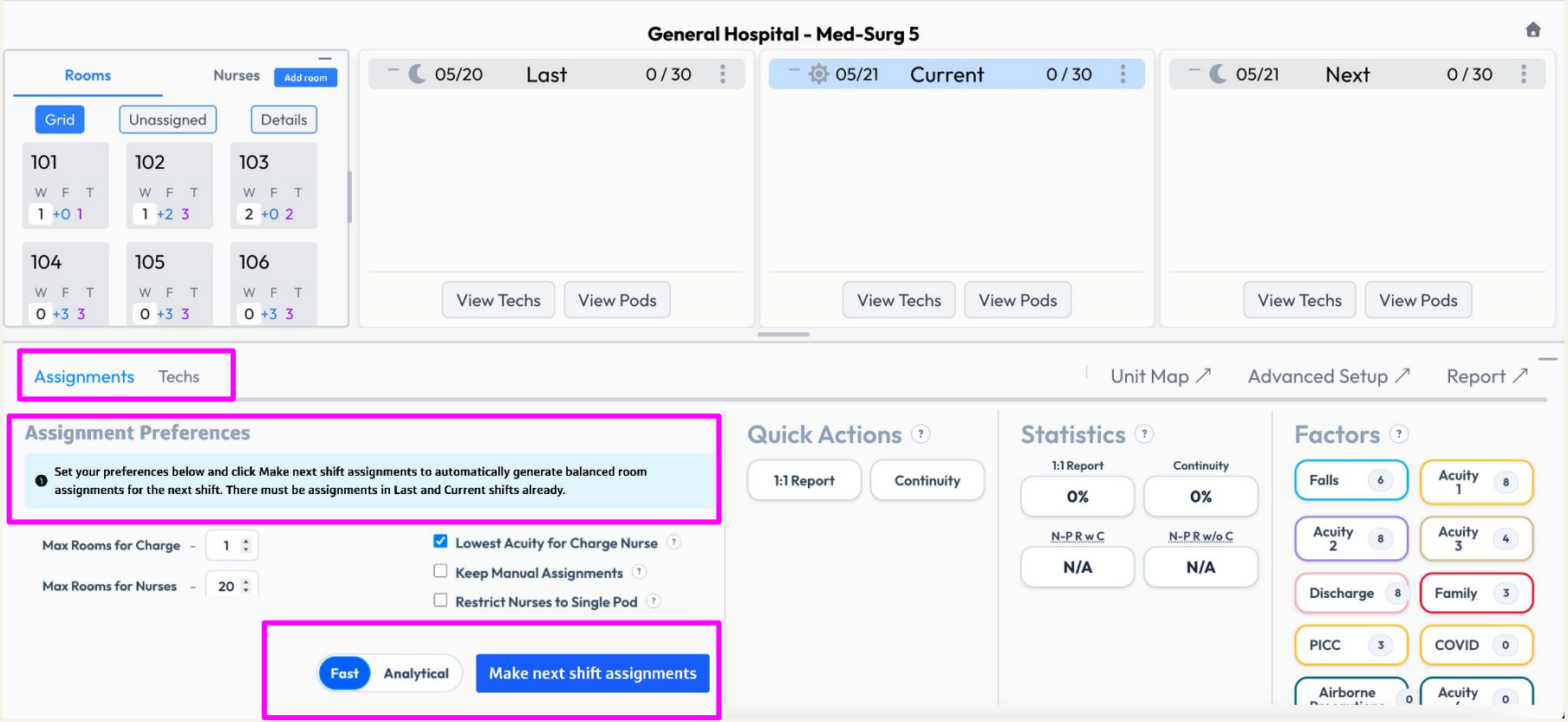
ISSUE

Users were struggling to find the Assignments tab in the bottom panel and needed much guidance to find it at all.

RECOMMENDATION

Have the Assignments tab be on the bottom panel at first view instead of the nurses. Provide guidance on what the Assignments tab is for and point them to the next step (add nurse to shift, add room, etc).

Provide guidance to the Assignments tab and make it easier to find



Allow dragging and dropping a nurse to a room to match the mental model of how nurses think.

ISSUE

The nurse's mental model is to drag a nurse into a room because they are adding a nurse to a room, not adding a room to a nurse.

RECOMMENDATION

Allow the user to drag and drop the nurse into a room, similar to how they can drag a nurse into a shift. Then the room will update with the nurse name and the room will be added to the nurse.

Adopt a "Primary/Secondary" button pattern to provide clearer distinction between different actions.

ISSUE

Because the button styling is very similar across the platform, it is difficult to distinguish between what actions are most important.

RECOMMENDATION

The main action should be a filled, high-contrast button, while the alternative action ("Advanced Setup") should be a secondary, ghost/outlined, or text-link button. This guides the user to the intended path immediately.

Add a tooltip to the workload score explaining how it is being used on the platform.

ISSUE

Users have a general idea of what the workload score means, but are not 100% certain of how and where it is being used.

RECOMMENDATION

Add an icon tooltip next to the workload score so that when they hover, it explains what exactly the workload score is and how they can use it to help in their assignments, and where it appears in the assignments page.

Transition hover actions to be nested within an overflow icon to prevent missing actions on hover.

ISSUE

Users are having difficulty setting a nurse to charge nurse because it's difficult to see that button appearing on hover.

RECOMMENDATION

Add an overflow icon to the nurses similar to the rooms and add a button under it that says "Charge nurse." Or make the C icon more noticeable on hover (but this may result in clicking it accidentally, not knowing what it does).

Appendix

Reference Materials

1. Kickoff Deck
2. Kickoff Questions
3. Dissertation
4. Nurse Feedback
5. Research Synthesis
6. How Nurse Scheduling Works
7. Heuristic Findings Deck
8. Google Sheet of Heuristic Results
9. Heuristic Flow
10. Heuristic Flow of Nurse Handoff
11. UX Toolkit